Rainwater-harvesting Hospital in Shyamnagar: A Climate-conscious Design

Team Number: 11

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Introduction

The Friendship hospital, located in Shyamnagar village in southwest Bangladesh's Satkhira district – an area vulnerable to frequent storms and rising sea levels, is recently awarded 'the world's best new building' by the Royal Institute of British Architects (RIBA) for its climate-conscious behaviour along with care and humanity. Kashef Chowdhury, the chief architect of the project, used water as its chief starting point by conducting rainwater harvesting keeping the villagers in mind. Rainwater harvesting is an alternative water supply method, with many environmental benefits, which produces drinking water. Truly, this hospital is a celebration of a building dedicated to humans and nature.

Background and Challenges

Located in Shyamnagar in the southern region of Bangladesh, Friendship Hospital was designed to meld with the water-laden local environment, which is filled with shrimp fisheries and crisscrossed by rivers. As a result, the structure faces significant climate, economic, environmental, and material availability challenges. Some of the vital issues are:

- Situated in a fragile and dynamic environment, directly impacted, without protection, by rising sea levels.
- The area was heavily affected by a significant cyclone Sidr (Category IV) in 2007.
- Built for a tight budget of just under $2m.
- Needed to withstand demanding environmental conditions due to rising sea levels impacted by climate change.
- The ample water supply surrounding the structure cannot be easily used for any purpose because of its high salinity level.
- Surrounded by former grain fields that now sit underwater due to sea-level rise high salinity.
- Unbearably hot summers required high energy-consuming air conditioning.

Methodology

- A series of courtyards bring natural light and sun, and the central canal collects rainwater from every roof and courtyard and leads to storage tanks at either end of the site.
- Buildings are set at angles with deep outdoor corridors, framed with colonnades, offering cross ventilation and encouraging natural cooling.
- Blocks are angled to reduce the need for air conditioning as it takes advantage of the prevailing wind direction.

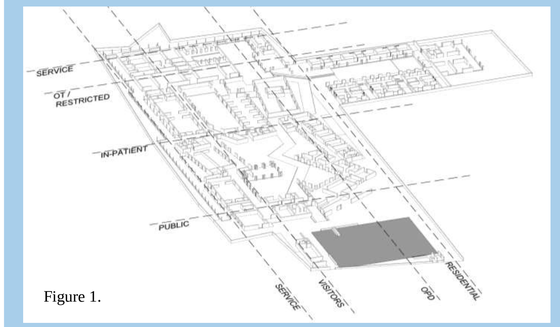


Figure 1.

- A brick water tower providing a civic focal point is located at the centre, which acts as a campanile.
- Locally made bricks are used for low cost and resilient qualities to provide longevity. Openings in the brickwork offer private shaded areas and natural cooling.
- A service corridor frees the central route for patients and medical staff, allowing clear views from patients' beds.
- Inpatient and outpatient departments are separated across shared courtyards.



Figure 2.

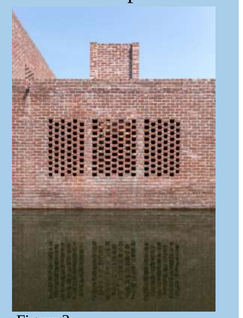


Figure 3.

Results

Friendship Hospital project offers both health benefits, social benefits, and environmental benefits. This hospital is a celebration of a building dedicated to humans and nature.

Health and Social Benefits:

- Provides vital medical facilities to the rural community
- Utilize rainwater and provide usable water to its locality for drinking and irrigation.
- The canal that runs through the hospital helps with microclimatic cooling.
- Provides gross ventilation which bounces daylight and allows prevailing wind back inside.
- Its well-ventilated aesthetic interior and soothing surrounding improve patients' mental health.



Figure 4.



Figure 5.

Environmental Benefits:

- Reduces wastage of water and energy.
- Reduce emission to air.
- Enhance existing ecology.
- Protects existing natural spaces
- Protects biodiversity and ecosystem
- Conserves natural resources
- Improves air and water quality

Conclusion

Lauded for the care and humanity at the heart of its design, the building responds innovatively to the challenging environmental conditions of the Bengal, a fragile and dynamic environment, directly impacted, without protection, to rising sea levels. It is a demonstration of how beautiful architecture can be achieved through good design when working with a relatively modest budget and with difficult contextual constraints.

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